Standard form

1

a $2x - y = 5$ **b** $2x - 5y = -20$ **c** $6x - y = 5$ **d** $6x - 5y = 30$

e $x - 6y = 0$ **f** $5x - y = -5$

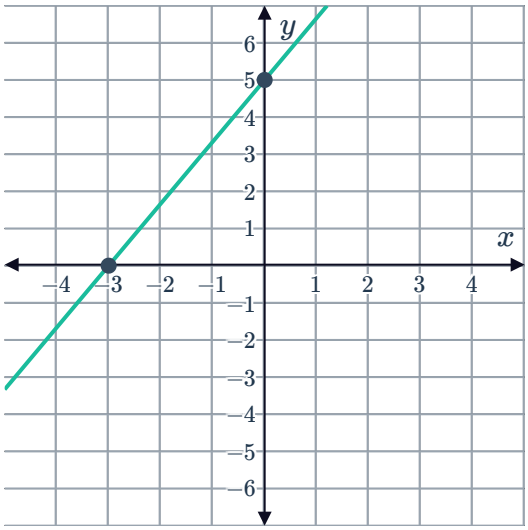
2 Step 3

3

a

i $x = -3$ **ii** $y = 5$

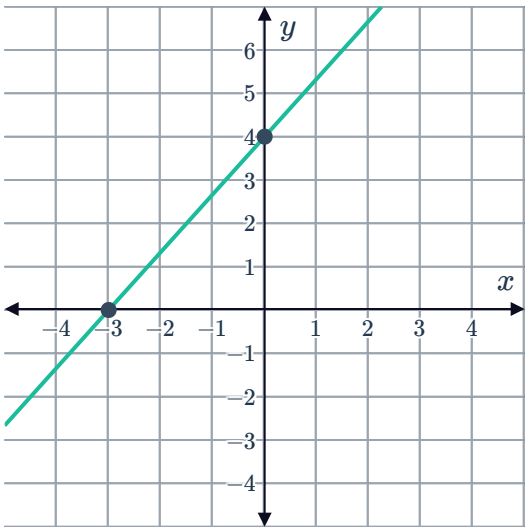
iii



b

i $x = -3$ **ii** $y = 4$

iii



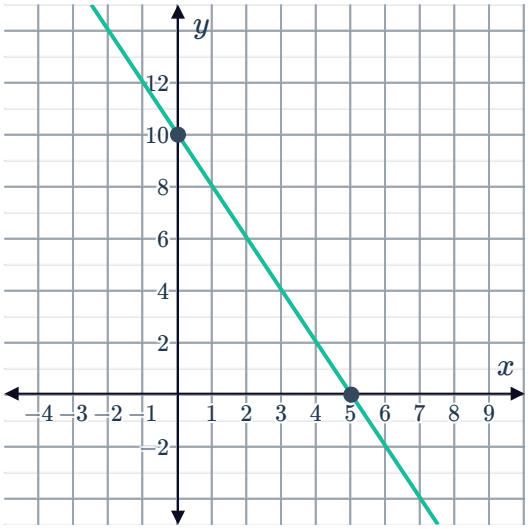
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c

i $x = 5$

ii $y = 10$

iii

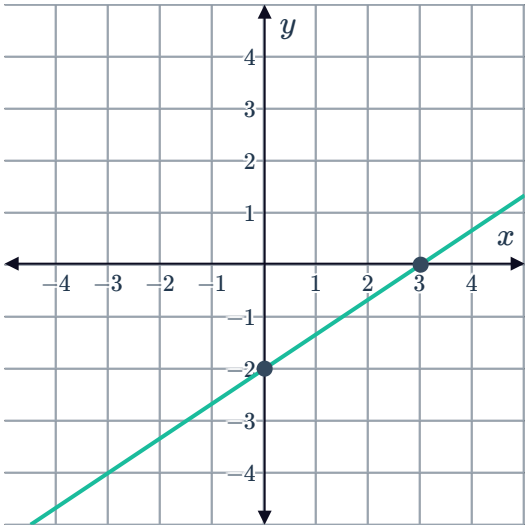


d

i $x = 3$

ii $y = -2$

iii

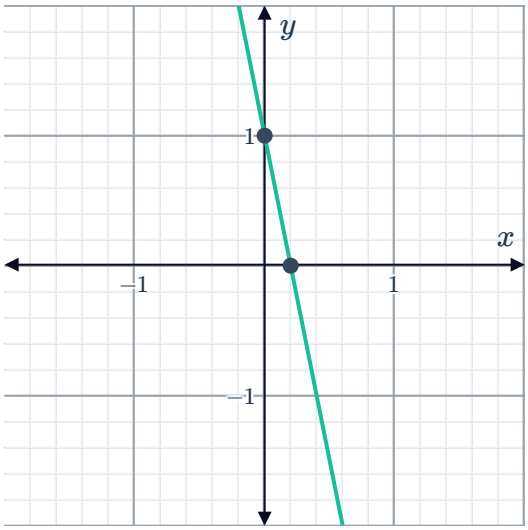


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e

i $x = \frac{1}{5}$ ii $y = 1$

iii



4

a

i $y = \frac{x}{3} - \frac{1}{3}$ ii $\frac{1}{3}$ iii $-\frac{1}{3}$

b

i $y = 6x + 8$ ii 6 iii 8

c

i $y = -\frac{2x}{5} + 7$ ii $-\frac{2}{5}$ iii 7

5

a

i $b = -\frac{76}{3}$ ii $7x - 3y = 76$

b

i $b = \frac{-59}{5}$ ii $10x - 5y = 59$

6

a $y = 4x + \frac{1}{5}$ **b** $20x - 5y = -1$ **c** $x = -\frac{1}{20}$

7

a $y = -\frac{4x}{3} - \frac{5}{3}$ **b** $y = -\frac{7x}{9} + \frac{1}{9}$ **c** $m\mathrm{A} = -\frac{4}{3}$ $m\mathrm{B} = -\frac{7}{9}$ **d** Line A

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8 $5x + 2y = 5$

9 $x - y = -3$

10

a No **b** Yes **c** No **d** Yes

11 $y = \frac{2}{5}$

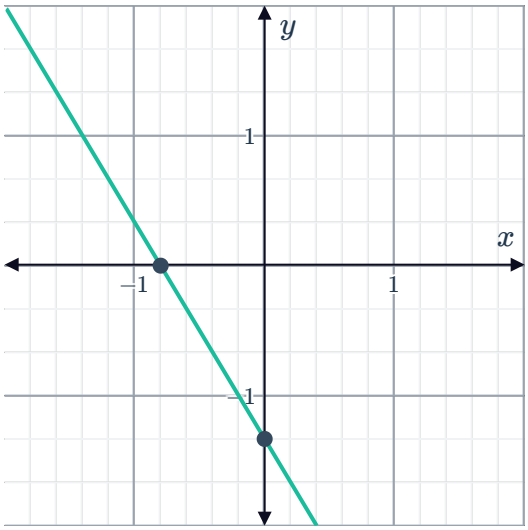
12

a $a = -5$ **b** $\frac{5}{3}$

13

a $5x + 3y = -4$

b



14

a 2 **b** -4 **c** $y = -4x + 2$ **d** $4x + y = 2$

15

a $k = 3$ **b** $x = 5$ **c** $y = 5$

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Additional questions

16

a $3x + 6y = 30$

b $x = 10$, so she could buy 10 green apples and no red apples for $\$15$.

c 6 green apples

d Domain$= [0, 10]$

17 $m = -\dfrac{A}{B}$, where $B \neq 0$

This could be helpful as we are often asked about rate of change in context and this quickly gives us that information. Also, it is a good check when we graph using intercepts to check that the slope is correct.

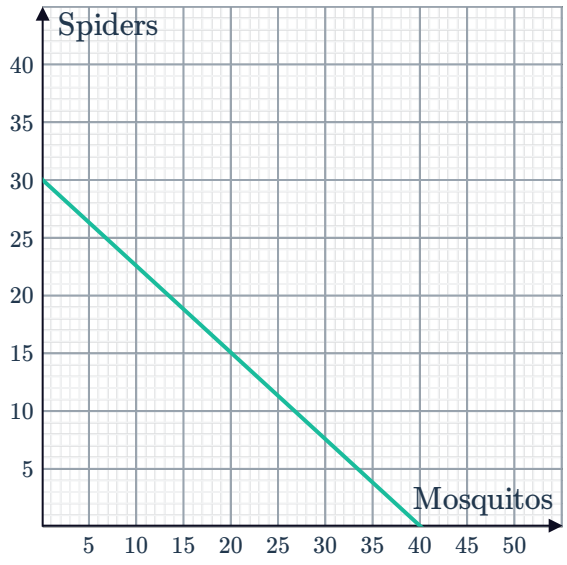
18 A scenario where we are creating a mixture or different portions of two substances and know the total amount. For example, if we are buying a mix of wheat and rye flour at the store and they have different prices and know that we need a total of 16 cups of flour.

19 They are similar as they both are forms of a linear equation. They are different because we can read off or easily calculate different key features from the equation.
A benefit of standard form is that we can easily calculate the x and y-intercepts by setting one of the variables to 0. A benefit of slope-intercept form is that we can read off the slope and y-intercept from the equation. From a graph we can also find the slope-intercept form more quickly as we can usually read off the slope and y-intercept from the graph.

20

a $6x + 8y = 240$

b



Domain $= [0, 48]$, the number of mosquitos between 0 and 48 mosquitos

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d We can only have whole numbers for the number of mosquitos and spiders, so no, every point on the line does not represent a solution, for example there are no valid points between $(0, 30)$ and $(4, 27)$.

21

a $3x - 5y = 30$

b The x-intercept is $x = 10$ which means if she sells 10 baby outfits, she will buy 0 toddler outfits.

c If we substitute $y = 15$ into our equation, we find that $x = 35$. This means that in order to buy 15 toddler outfits and make a profit of $\$30$, she would need to sell 35 baby outfits, but she only has 25, this is not enough. It is not possible to make her desired profit.

d Domain $x = [10, 25]$, because if she was to sell less than 10 baby outfits she would not make her desired profit, so she must sell at least 10 baby outfits and she only has 25 baby outfits, so that is the maxiumum value.

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